

Shopify Stock Data

Week 4 – Power BI EDA

Exploratory Data Analysis, Moving Averages & Interactive Dashboard

Prepared using	Power BI Desktop
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1. Calculate SMA (Simple Moving Average) for 20 Days and 50 Days Using DAX

Two moving-average measures were created for Shopify trading volume. The actual Power BI model uses the Date Table and Shopify_Stock table.

Volume 20-Day Simple Moving Average

```
Volume 20 SMA =  
VAR CurrentDate = MAX('Date Table'[Date])  
VAR PeriodDays = 20  
VAR DateInPeriod =  
DATESINPERIOD(  
'Date Table'[Date],  
CurrentDate,  
-PeriodDays,  
DAY  
)  
RETURN  
CALCULATE(  
AVERAGE(Shopify_Stock[Volume]),  
DateInPeriod  
)
```

Volume 50-Day Simple Moving Average

```
Volume50SMA =  
VAR CurrentDate = MAX('Date Table'[Date])  
VAR PeriodDays = 50  
VAR DateInPeriod =  
DATESINPERIOD(  
'Date Table'[Date],  
CurrentDate,  
-PeriodDays,  
DAY  
)  
RETURN  
CALCULATE(  
AVERAGE(Shopify_Stock[Volume]),  
DateInPeriod  
)
```

Moving Average Crossover Signal

```
MACrossoverSignal =  
IF(  
ISBLANK([Volume 20 SMA]) ||  
ISBLANK([Volume50SMA]),  
BLANK(),  
)
```

```

IF(
[Volume 20 SMA] > [Volume50SMA],
"Bullish (20 > 50)",
"Bearish (20 < 50)"
)
)

```

2. Visualizations

After creating the DAX measures, the report page was used to build interactive stock-price, volume, filtering, and parameter-selection visuals.

3. Interactive OHLC & Price Trend / Moving-Average Visual

A Line and Stacked Column Chart was configured with Date on the timeline and the stock-price / moving-average measures. The X-axis Zoom slider was enabled for timeline navigation. The moving-average measures were used together with Close Price for trend analysis.

4. Trading Volume & Volume 20 SMA Visual

A volume visual was created using Date on the X-axis and Trading Volume together with Volume 20 SMA. The visual allows trading-volume activity to be compared with its 20-day moving average. The optional conditional colour formatting from the instructions was not applied.

5. Timeline Slicer & Dynamic Metric Selector

- **Date Range Slicer:** A Date Table[Date] slicer was created and configured in Between mode.
- **Parameter Switch:** A Fields parameter named Price Metric Selector was created with Close Price, High Price, Low Price and Open Price.
- **Dashboard interaction:** The parameter selector is visible on Page 2 and can be used to switch between the four price measures.

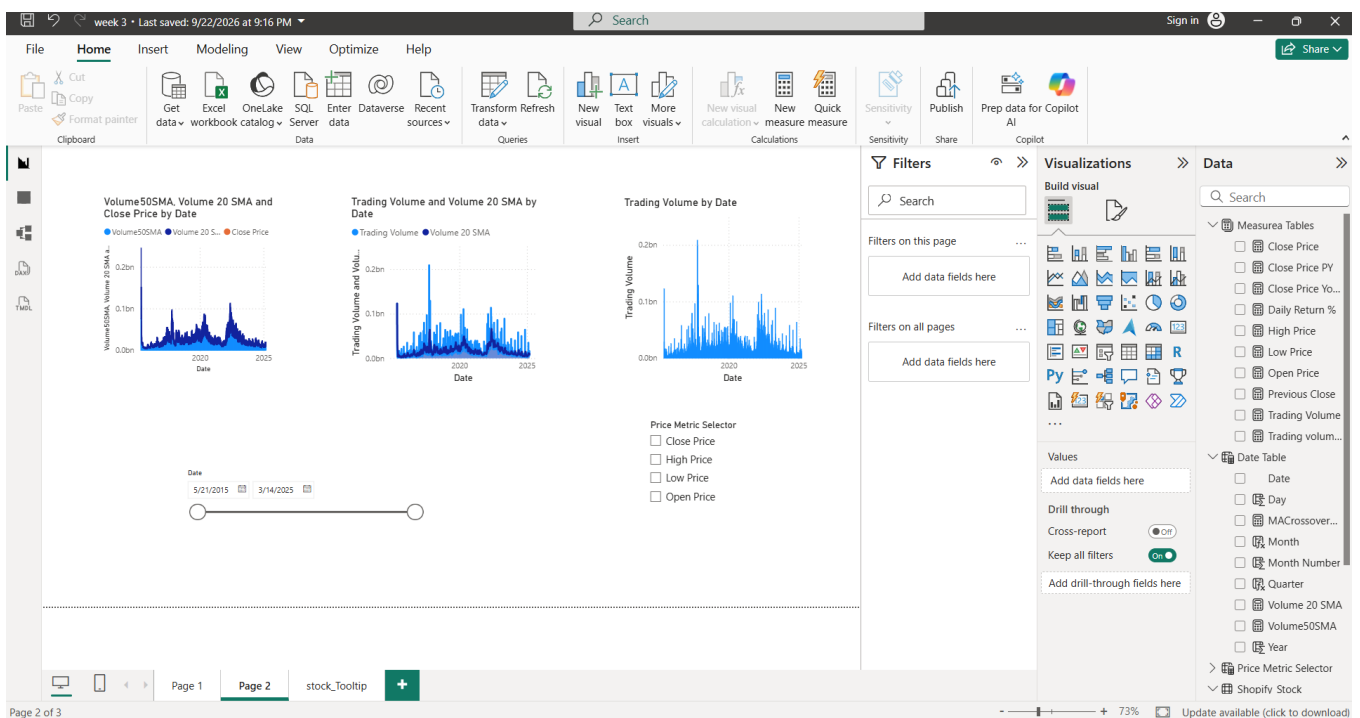


Figure 1: Week 4 Page 2 showing moving-average/volume visuals, Date Between slicer and Price Metric Selector.

6. Custom Tooltip Overlay Page

A separate report page named Stock_Tooltip was created and configured as a tooltip page. KPI cards were added to provide selected-date stock details. The completed tooltip page visibly contains Open Price, High Price, Low Price, Close Price and Trading Volume.

The Daily Change % card was not included in the final completed tooltip screenshot, so it is not claimed as completed in this report.

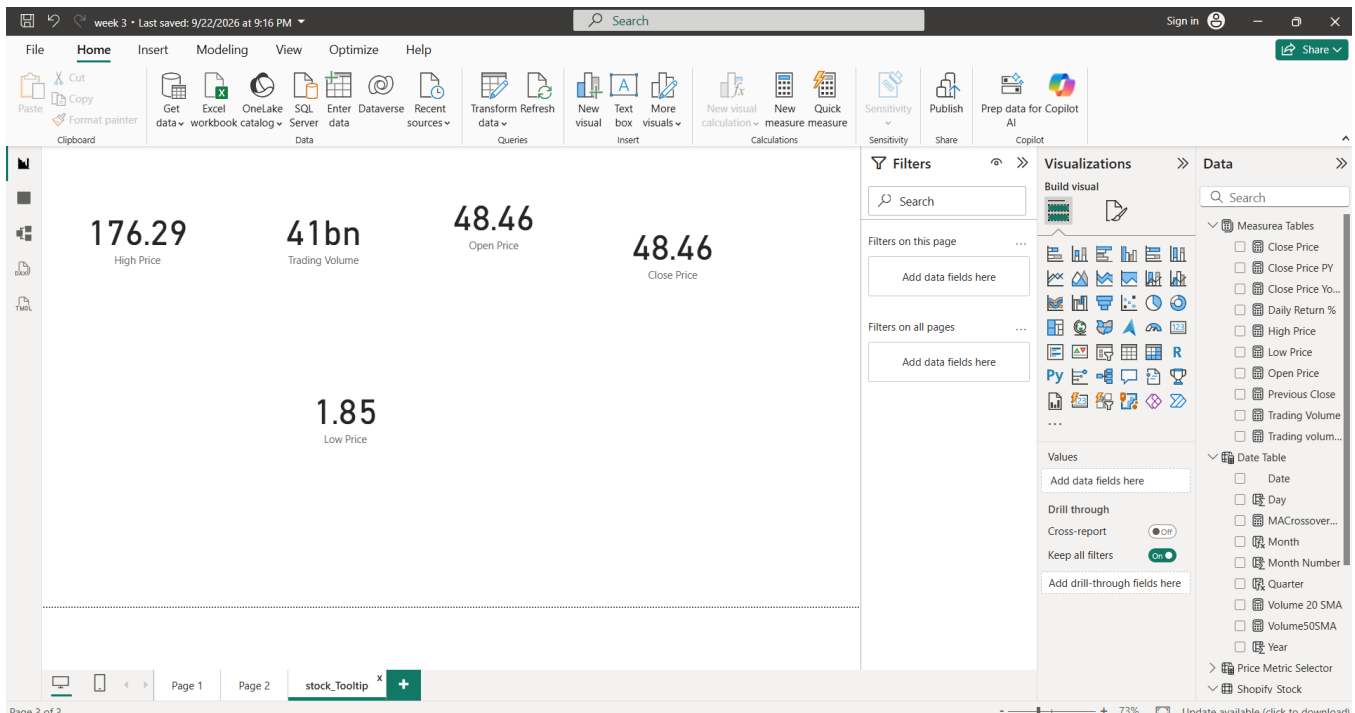


Figure 2: Stock_Tooltip page showing the completed KPI cards.

7. Final Output

The Week 4 Power BI EDA report includes 20-day and 50-day volume moving averages, a moving-average crossover signal, stock/volume trend visuals, a Date Between slicer, a dynamic Price Metric Selector, and a custom Stock_Tooltip page with the completed KPI cards.

8. Conclusion

During Week 4, Shopify stock data was analyzed using DAX-based moving averages and interactive Power BI features. The dashboard was extended with moving-average analysis, timeline filtering, dynamic price selection, and a custom tooltip page to support clearer exploration of stock performance and trading activity.